Data Quality Report

AeroGrid | Source: live Neon warehouse | UTC: 2026-11-13T21:11:41.876426+00:00

Audit summary

DQ run: 11; critical FAIL checks: 0; warning checks: 4. Refer to each check below.

Check catalog results

Check ID	Severity	Status	Observed / Expected
DQ-A-001	CRITICAL	PASS	0 / 0
DQ-A-002	CRITICAL	PASS	0 / 0
DQ-A-007	CRITICAL	PASS	0 / 0
DQ-A-008	WARNING	WARN	14 / 0
DQ-B-001	CRITICAL	PASS	0 / 0
DQ-B-003	CRITICAL	PASS	0 / 0
DQ-B-004	CRITICAL	PASS	0 / 0
DQ-C-001	CRITICAL	PASS	0 / 0
DQ-C-003	WARNING	WARN	27479 / 0
DQ-C-005	CRITICAL	PASS	0 / 0
DQ-C-006	CRITICAL	PASS	0 / 0
DQ-C-007	WARNING	PASS	0 / 0
DQ-C-008	WARNING	WARN	27479 / 0
DQ-G-001	CRITICAL	PASS	0 / 0
DQ-G-002	WARNING	PASS	0 / 0
DQ-G-003	WARNING	PASS	0 / 0
DQ-G-004	WARNING	PASS	0 / 0
DQ-I-001	CRITICAL	PASS	0 / 0
DQ-I-002	CRITICAL	PASS	0 / 0
DQ-I-003	CRITICAL	PASS	0 / 0
DQ-I-004	CRITICAL	PASS	0 / 0
DQ-I-005	CRITICAL	PASS	0 / 0
DQ-I-006	CRITICAL	PASS	0 / 0
DQ-I-008	CRITICAL	PASS	0 / 0
DQ-I-009	CRITICAL	PASS	0 / 0
DQ-I-010	CRITICAL	PASS	0 / 0
DQ-R-001	CRITICAL	PASS	0 / 0
DQ-R-002	CRITICAL	PASS	0 / 0
DQ-R-004	CRITICAL	PASS	0 / 0
DQ-R-005	CRITICAL	PASS	0 / 0
DQ-R-006	CRITICAL	PASS	0 / 0

Check ID	Severity	Status	Observed / Expected
DQ-T-003	WARNING	WARN	1 / 0
DQ-U-001	CRITICAL	PASS	0 / 0
DQ-U-002	CRITICAL	PASS	0 / 0
DQ-U-003	CRITICAL	PASS	0 / 0
DQ-U-004	CRITICAL	PASS	0 / 0

Open DQ warnings explained

Check	Observed	Meaning / source limitation
DQ-A-008	14	Expected invalid-row count: 0
DQ-C-003	27479	Unknown weather-key assignments; no matched real METAR weather observation is linked to these tracking facts.
DQ-C-008	27479	Source on_ground is NULL; WP-07 airborne eligibility is inferred and does not replace the observed field.
DQ-T-003	1	One warning indicator for missing real restriction snapshots; this is NOT one missing flight or one restriction event.

Source coverage (loaded vs not loaded)

Source	Staging table	Accepted rows	Status
FAA restrictions	stg_restriction_snapshot	0	NO DATA LOADED
OpenSky (optional, D-013)	stg_opensky_state	0	NO DATA LOADED
AWC/IEM METAR	stg_weather_metar	400	LOADED
FAA airspace zones	stg_airspace_zone	252	LOADED
Nightjar / nominal-io DJI	stg_uas_position	22602	LOADED
Tartan ADS-B (KBTP)	stg_adsb_position	27479	LOADED

Coverage meaning

NO DATA LOADED means the source was validated but never ingested. It does not mean zero events occurred. OpenSky is optional under D-013 and has no ingestion path. See docs/13_accepted_data_limitations.md.

Real warehouse row counts

Schema.table	Rows
project_staging.stg_adsb_position	27479
project_staging.stg_reject	578
project_staging.stg_airspace_zone	252
project_staging.stg_restriction_snapshot	0
project_dw.dim_time	335

Schema.table	Rows
project_dw.dim_site	5
project_dw.dim_vehicle	159
project_dw.dim_spatial_grid	3659
project_dw.dim_weather	401
project_dw.dim_airspace_zone	253
project_dw.fact_vehicle_tracking	27479
project_dw.fact_flight_journey	178
project_dw.fact_close_encounter	0
project_dw.fact_airspace_occupancy	1754
project_dw.fact_restriction_activation	0

Unknown dimension members

Dimension totals include the mandatory -1 unknown member. For example, a one-row dim_weather does NOT imply one measured weather observation, and a one-row dim_airspace_zone does NOT imply one real FAA restriction.

Capacity

Actual PostgreSQL database size: 75,022,336 bytes. Object-storage usage is not available from the PostgreSQL connection; do not equate manifest bytes with current bucket usage.

Encounter and restriction limitations

The WP07-v1 encounter algorithm requires identical timestamps. Near-time diagnostics were separate analytical SQL, not the production algorithm. No historical 2022 TFRs are inferred from 2026 FAA services. Zero derived rows may be legitimate source coverage, not evidence of a missing SQL table. Read the two zero-row views together with the source coverage table above and with docs/13_accepted_data_limitations.md: an unloaded source is not a measured zero.

Decision rule

Any stored CRITICAL FAIL blocks trusted publication even if this PDF can be generated. A green unit-test run alone does not establish data quality.

Evidence and reproducibility

Latest ETL run ID: 11, mode=weekly, started=2026-11-13 20:34:19.325115+00:00, status=SUCCESS, version=wp11-v1. DQ check IDs map to Phase2/etl/dq.py and Phase2/docs/06_data_quality_and_reports.md. Verify the source manifests and rejected-row counts directly in Neon.